

Algebra 1
Unit 1
Lesson 7 Homework

Name **Answer Key** _____
Date _____
Period _____

1. Write the expression $64^{\frac{3}{2}}$ in two different equivalent forms.

Expression 1	Expression 2
$(64^3)^{\frac{1}{2}}$	$\sqrt{64^3}$

2. The value of $64^{\frac{3}{2}}$ is

- ☐ 16
- ☐ 42
- ☐ 83
- ☐ 512

For questions 3-5, write an equivalent radical expression and then find the value of the expression.

3. $(-216)^{\frac{2}{3}} = \sqrt[3]{(-216)^2} = 36$

4. $-\left(\frac{32}{243}\right)^{\frac{4}{5}} = -\sqrt[5]{\left(\frac{32}{243}\right)^4} = -\frac{16}{81}$

5. $(144)^{-\frac{1}{2}} = \frac{1}{\sqrt{144}} = \frac{1}{12}$

Evaluate each expression in questions 6-10 without a calculator.

$$6. \quad -(2^{-2}) \left(81^{-\frac{1}{2}} \right) = - \left(\frac{1}{2^2} \right) \left(\frac{1}{\sqrt{81}} \right) = - \left(\frac{1}{4} \right) \left(\frac{1}{9} \right) = \frac{-1}{36}$$

$$7. \quad \left(32^{\frac{1}{5}} \right)^3 \cdot 8^{-1} = 2^3 \left(\frac{1}{8} \right) = \frac{8}{8} = 1$$

$$8. \quad - \left(\frac{16}{64} \right)^{-\frac{3}{2}} = - \left(\frac{64}{16} \right)^{\frac{3}{2}} = - \sqrt{\left(\frac{64}{16} \right)^3} = - \sqrt{4^3} = -8$$

$$9. \quad (-1^{10}) \cdot \left(-125^{\frac{1}{3}} \right) = (1)(-5) = -5$$

$$10. \quad \left(\frac{7}{21} \right)^{-3} \cdot (3^{-2}) = \left(\frac{21}{7} \right)^3 \cdot \left(\frac{1}{3^2} \right) = (27) \left(\frac{1}{9} \right) = 3$$